

CSCI 5502

Data Mining

FORMAT	CATEGORY	CREDITS
On-Campus	Core	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 1 – CS Pathway (on-campus)	Machine Learning Engineer (71.0 / 100)	55.9 / 100
SPECIALIZATION Computer Science Pathway		

56

CURRENT AVG SCORE
across 6 job roles

98

POTENTIAL SCORE
with recommended additions

+60

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces basic data mining concepts and techniques for discovering interesting patterns hidden in large-scale data sets, focusing on issues relating to effectiveness and efficiency. Topics covered include data preprocessing, data warehouse, association, classification, clustering, and mining specific data types such as time-series, social networks, multimedia, and Web data.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	71.0	100.0	+29.0	MEDIUM
ML Engineer	68.9	95.1	+26.2	MEDIUM
Data Scientist	62.1	100.0	+37.9	HIGH
MLOps Engineer	48.7	100.0	+51.3	HIGH
AI Engineer	48.2	98.3	+50.1	HIGH
Applied Scientist	36.3	96.5	+60.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Model Deployment & Serving

2 of 6 roles

production

Experiment Tracking

1 of 6 roles

tracking

Business Communication

1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2

MEDIUM

Consider Model Deployment & Serving content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

3

LOW

Optional: Experiment Tracking — would broaden appeal to one additional role. Candidate terms: tracking.

4

LOW

Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

71.0 → 100.0 (+29.0 pts)

engineering

learning

analytics

tools

solutions

business

mobile

time

ML Engineer

68.9 → 95.1 (+26.2 pts)

discovery

learning

machine

systems

models

production

engineering

starting

Data Scientist

62.1 → 100.0 (+37.9 pts)

characteristics

analytics

evolution

build

engineering

MLOps Engineer

48.7 → 100.0 (+51.3 pts)

infrastructure

detection

production

text

systems

learning

machine

tracking

AI Engineer

48.2 → 98.3 (+50.1 pts)

following

engineering

models

learning

infrastructure

build

software

systems

Applied Scientist

36.3 → 96.5 (+60.2 pts)

discovery

scientific

research

literature

models

learning

systems

industry

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.